In October 1967—commemorating the tenth anniversary of *Sputnik 1*—the Soviet *Venera 4* spacecraft dropped an entry capsule into the clouds of Venus. It returned data from the hot lower atmosphere, but did not survive to the surface. One day later, the United States spacecraft *Mariner 5* flew by Venus, its radio transmission to Earth skimming the atmosphere at progressively greater depths. The rate of fading of the signal gave information about atmospheric temperatures. Although there seemed to be some discrepancies (later resolved) between the two sets of spacecraft data, both clearly indicated that the surface of Venus is very hot.

Since then a progression of Soviet *Venera* spacecraft and one cluster of American spacecraft from the *Pioneer 12* mission have entered the deep atmosphere or landed on the surface and measured directly—essentially by sticking out a thermometer—the surface and near-surface temperatures. They turn out to be about 470°C, almost 900°F. When such factors as calibration errors of terrestrial radio telescopes and surface emissivity are taken into account, the old radio observations and the new direct spacecraft measurements turn out to be in good accord.

Early Soviet landers were designed for an atmosphere somewhat like our own. They were crushed by the high pressures like a tin can in the grasp of a champion arm wrestler, or a World War II submarine in the Tonga Trench. Thereafter, Soviet Venus entry vehicles were heavily reinforced, like modern submarines, and successfully landed on the searing surface. When it became clear how deep the atmosphere is and how thick the clouds, Soviet designers became concerned that the surface might be pitch-black. *Veneras 9* and *10* were equipped with floodlights. They proved unnecessary. A few percent of the sunlight that falls on the top of the clouds makes it through to the surface, and Venus is about as bright as on a cloudy day on Earth.

The resistance to the idea of a hot surface on Venus can, I suppose, be attributed to our reluctance to abandon the notion that the nearest planet is hospitable for life, for future exploration, and perhaps even, in the longer term, for human settlement. As it turns out there are no Carboniferous swamps, no global oil or seltzer